SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	ERROR 303
Project / Product Name	StockWise
Final Live Deployment URL	https://error303v2.vercel.app/
Codebase Repository URL	https://github.com/timlai0402/ERROR303-2.0.git
Name (Leader)	Timothy Lai Yong Fong
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2. Project Description

[Maximum 250 words]

StockWise helps small and medium-sized businesses manage inventory through a simple, mobile-friendly web application. It addresses repetitive stock entry, inconsistent batch rotation and late reordering by bringing these tasks into one workflow. Staff can scan a product barcode or QR code, record stock movements and follow first-in, first-out (FIFO) batch-picking instructions. Snap Delivery Note uses ChatGPT through KrackkedDev to extract supplier document line items and suggest product matches. Users review and correct the results before confirming stock-in batches. The dashboard estimates days until stock runs out using the previous 14 days of recorded usage, checks supplier lead times and highlights low-stock items. Editable reorder templates open in WhatsApp or email for the user to send. These forecasts and message templates are calculated locally rather than generated by an AI model. The prototype includes inventory search, movement history, staff attribution, statistics, configurable alerts and optional Supabase synchronization across devices. Bundled sample notes support a demonstration when live AI is unavailable. StockWise targets more consistent stock rotation and better-informed replenishment, with potential alignment to SDG 12 through reduced avoidable inventory waste. Environmental savings remain a pilot-validation objective, not a measured outcome. The next stage is to validate usability and operational impact with SME users and strengthen authentication, access controls and data handling for production.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Component	Proposed Technology	Purpose
Frontend	React 18, Vite, React Router, HTML/CSS/JavaScript	Responsive inventory screens, routing and input workflows.
Backend	Node.js and Express	Receives document images at /api/extract-delivery-note; calls the configured AI service using server-side credentials.
GenAI	ChatGPT accessed through KrackkedDev; underlying model version unspecified	Extracts supplier/document details, quantities and product matches. Staff review output before saving.
Database	Supabase PostgreSQL, supabase-js and Realtime	app_users, suppliers, products, batches, movements and settings; direct browser access and live updates when configured. Browser storage is the fallback.
Inventory logic	Custom JavaScript in src/lib/inventory.js	Oldest-batch FIFO allocation, 14-day average-usage forecasts, lead-time alerts and local similarity matching.
Scanning and reporting	html5-qrcode, Recharts and lucide-react	Camera-based scanning, inventory charts and interface icons. Reorders open WhatsApp/mailto links.
Runtime and deployment	Vite build; Express production serving; GitHub Pages static hosting declared	Browser -> Express -> AI -> editable review -> inventory storage. Browser <-> Supabase handles synchronization. Static hosting requires a separate API backend; final deployment is pending confirmation.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

The current prototype implements product management, stock-in batches, FIFO stock-out allocation, barcode/QR scanning, low-stock alerts, movement history and statistics. Stock movement records include the selected staff member and timestamp. Snap Delivery Note integrates image compression, a server-side vision API call, structured extraction, product matching and an editable review step. Confirming the reviewed rows creates inventory batches. Three bundled sample notes use fixed extraction results when the live API is unavailable, allowing the workflow to be demonstrated transparently. Supabase integration provides shared persistence and Realtime updates when configured, while browser storage supports a local demonstration. The backend can serve the production frontend build and the AI endpoint. The final live deployment and API host still require confirmation. Implemented problem-solving measures include keeping the AI key on the server, reducing image size before upload, allowing corrections to extracted quantities and matches, and using local similarity matching when an extracted match cannot be resolved. Forecasting uses recorded stock-out data and supplier lead times; reorder messages are editable templates. A deterministic synthetic check of the current inventory module confirmed that a 25-unit withdrawal takes 20 units from the oldest batch and five from the next batch. A separate forecast check returned five days of stock and a 120-unit reorder suggestion for the inputs shown in Section 4.1. These statements describe the current code. The team must confirm which features were completed during the official 48-hour sprint and identify any changes from its Phase 1 submission before signing.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	SDG 12, particularly waste prevention through improved stock rotation, is the intended alignment. Proposed pilot targets: reduce median delivery-note entry time by 30%, reduce stock-record discrepancies by 20%, and achieve 100% oldest-available-batch allocation in controlled FIFO cases.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	Executed the existing allocate FIFO and forecast functions with fixed synthetic inputs dated 1 November 2026. FIFO allocated 20 units from the oldest batch and five from the next; the separate forecast returned 10 units/day, five days remaining, watch status and 120 units suggested. For a pilot, compare at least 30 matched delivery notes using manual entry and StockWise, recording time, corrections and final accuracy. Track inventory discrepancies, stockouts and discarded units over a defined baseline and pilot period. No pilot results are claimed.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Small retailers, offices and stockrooms could reduce repetitive entry and improve visibility of supplies. Human review keeps staff responsible for AI-assisted changes. A proposed SME pilot would assess ease of use, staff adoption and whether better stock rotation reduces avoidable disposal. Waste benefits depend on real

Metric Category	Your Response
	operating behavior and cannot be inferred from FIFO logic alone.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	The prototype uses a browser interface, an Express API and optional hosted PostgreSQL, allowing a small pilot without dedicated scanning hardware. A subscription per business or location is a proposed commercial model, with AI usage limits based on measured costs. Production readiness requires authentication, tenant isolation, atomic stock transactions, backups, monitoring and documented retention policies. Hosting costs, willingness to pay and final deployment access require validation.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Executed synthetic checks of src/lib/inventory.js These examples validate calculation behavior only; they do not establish real-world environmental savings. FIFO case: batch 0001 contains 20 units received on 1 October 2026; batch 0002 contains 30 units received on 20 October 2026. Request = 25 units. allocateFifo returns 20 from batch 0001 and 5 from batch 0002. Remaining quantities = 0 and 25. Conservation check: 50 - 25 = 25 units. Independent forecast case, before the withdrawal: stock S = 50; recorded stock-out over 14 days U = 140; supplier lead time L = 3 days; alert level A = 10. Average daily usage = U / 14 = 10. Days remaining = S / 10 = 5. Since 5 > L but 5 <= L + 4, status = watch. Target stock = 10 x (3 + 14) =

170. Suggested order = 5 x ceil(max(ceil(170 - 50), 10) / 5) = 120 units. These match the executed function outputs. Proposed pilot measurement: entry-time improvement (%) = 100 x (baseline median time - pilot median time) / baseline median time. Apply the same relative-change formula to inventory discrepancy rates, with consistent denominators and a nonzero baseline. Record actual discarded units separately; do not convert time saved or FIFO compliance into waste or CO2 savings without supporting measurements.

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

StockWise would begin with a small pilot involving SME retailers that currently use paper records or spreadsheets. Funding would support user interviews, onboarding, production security and measurement of entry time, inventory accuracy and stockouts. Initial targets would be treated as hypotheses and revised using pilot evidence. The first engineering priority is authenticated, tenant-isolated access with role permissions, row-level security and server-validated stock operations. Atomic database transactions would prevent simultaneous withdrawals from allocating the same stock. Backups, audit monitoring, retention controls and recovery testing would protect operational records. For higher document volumes, the AI backend would add request limits, queues, retries, structured-output validation and per-business usage budgets. A reviewed set of representative supplier documents would measure extraction accuracy and guide provider selection. Sensitive fields would be minimized before transmission where practical. A proposed subscription per business or location would cover hosting, support and an included AI allowance, with additional usage priced from measured costs. We would test willingness to pay before setting prices. Supplier partnerships and CSV import/export could support adoption and integration. Corporate, civic or university partners could provide pilot sites, implementation support and independent evaluation. Scaling would follow demonstrated user value, acceptable operating costs and verified data protection. Claims of reduced waste would be published only after a documented baseline comparison.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

- ☐ Yes
- ☑ No
- ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

- ☑ Yes
- ☐ No
- ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Official ChatGPT; Claude Code; Google Antigravity; Codex	Pre-trained AI model	NA
ChatGPT through KrackkedDev	Pre-trained API model	NA

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

- ☐ Yes
- ☑ No
- ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



1 Nov 2026